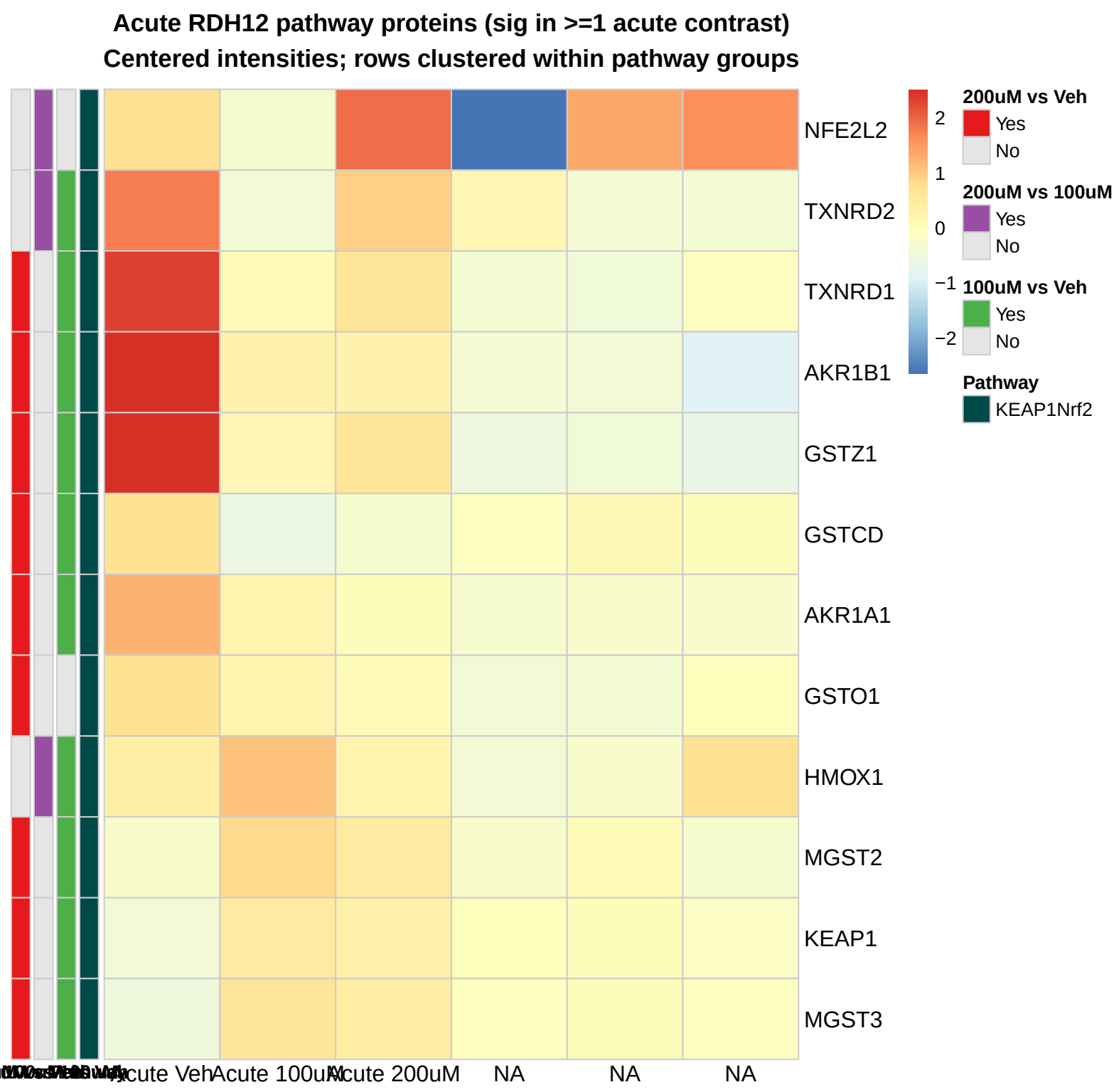
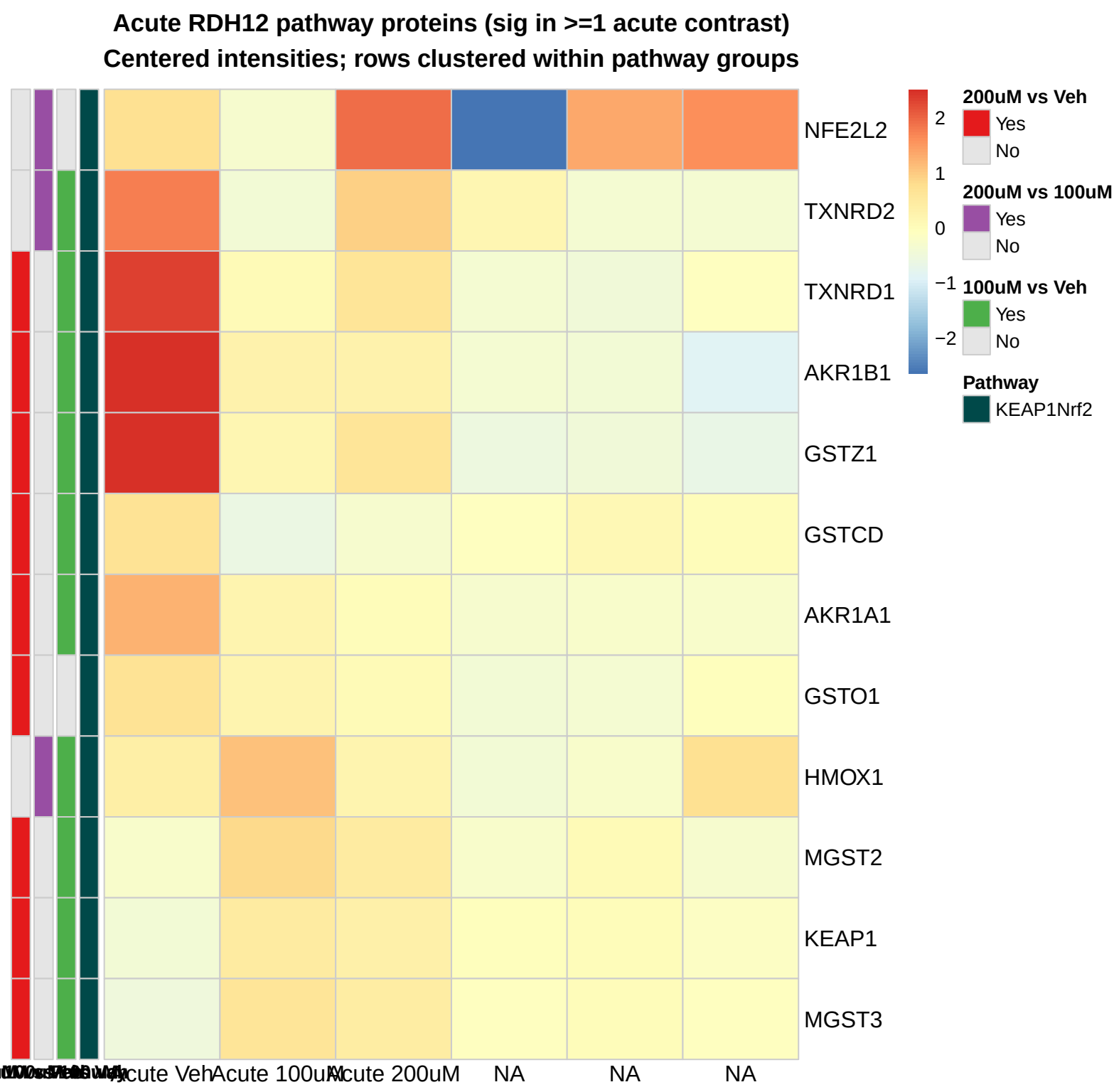
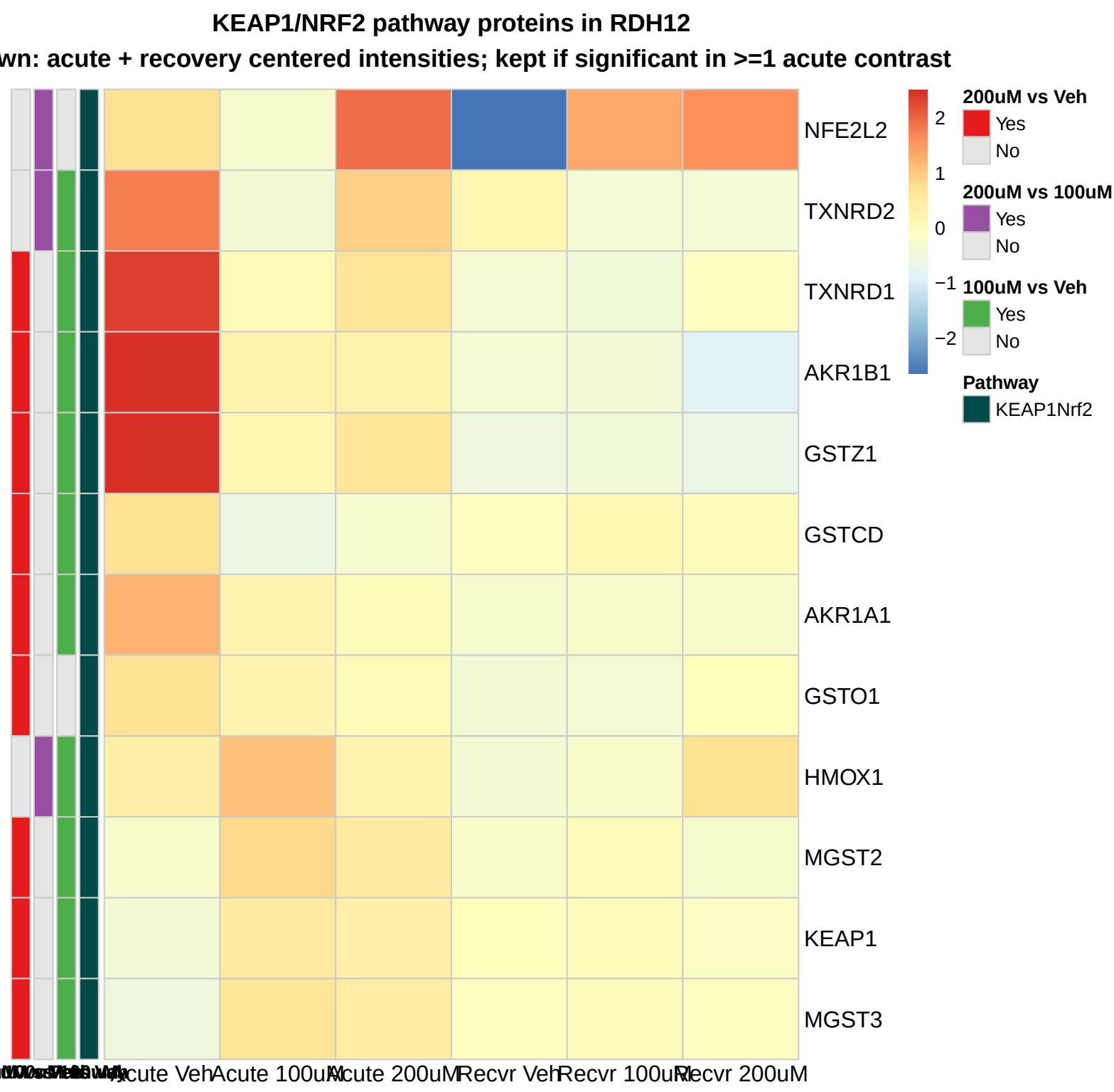
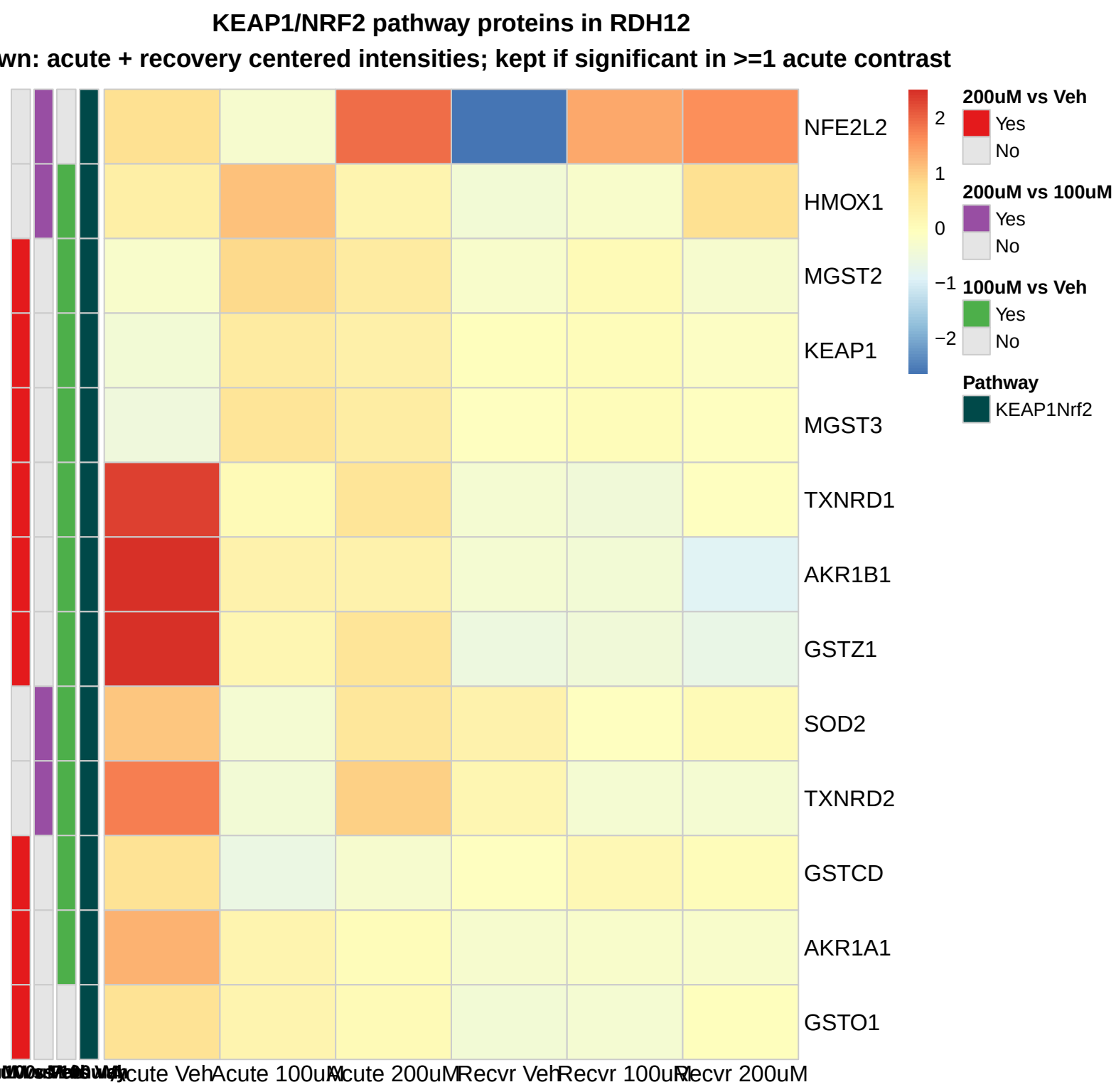
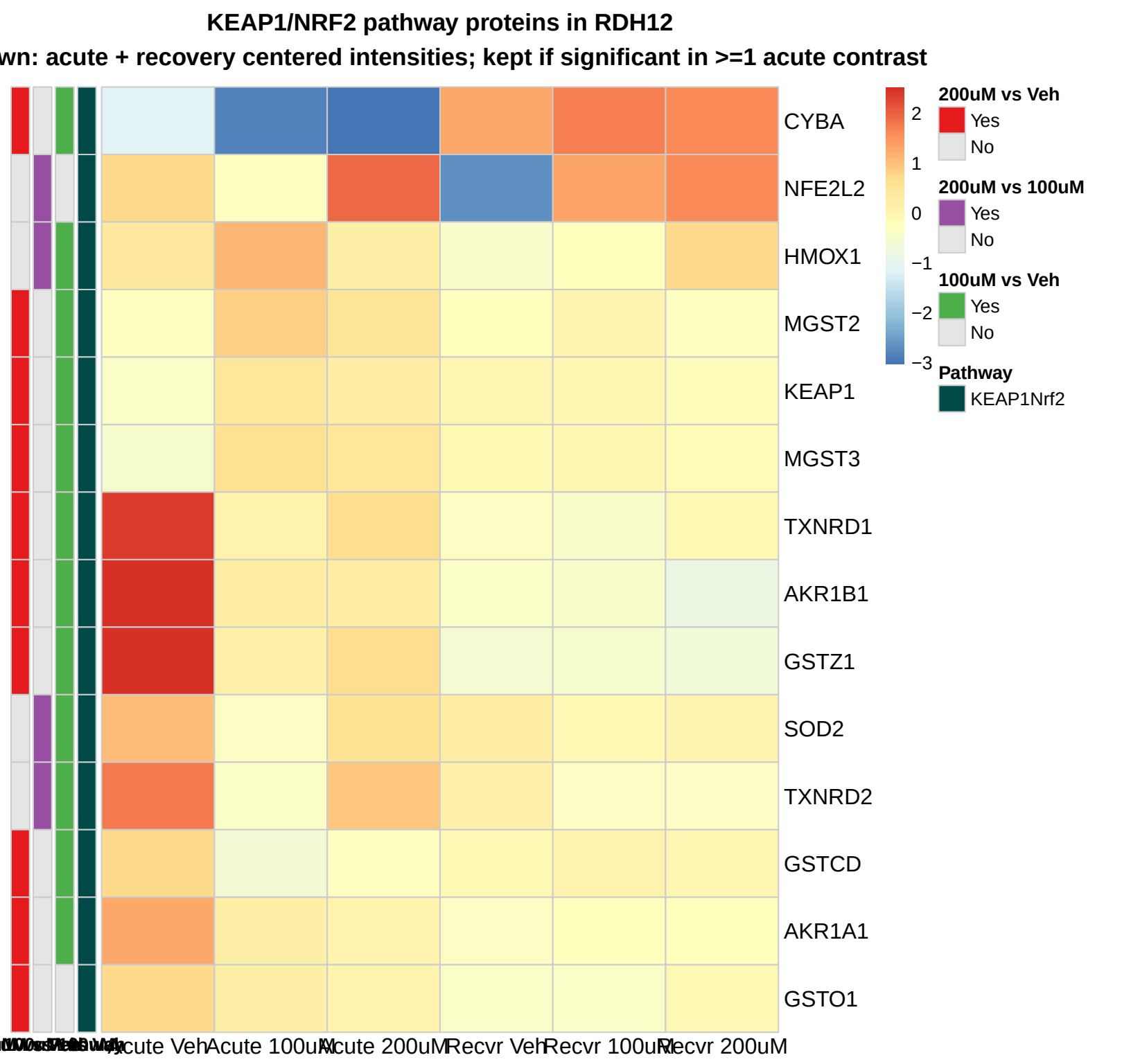
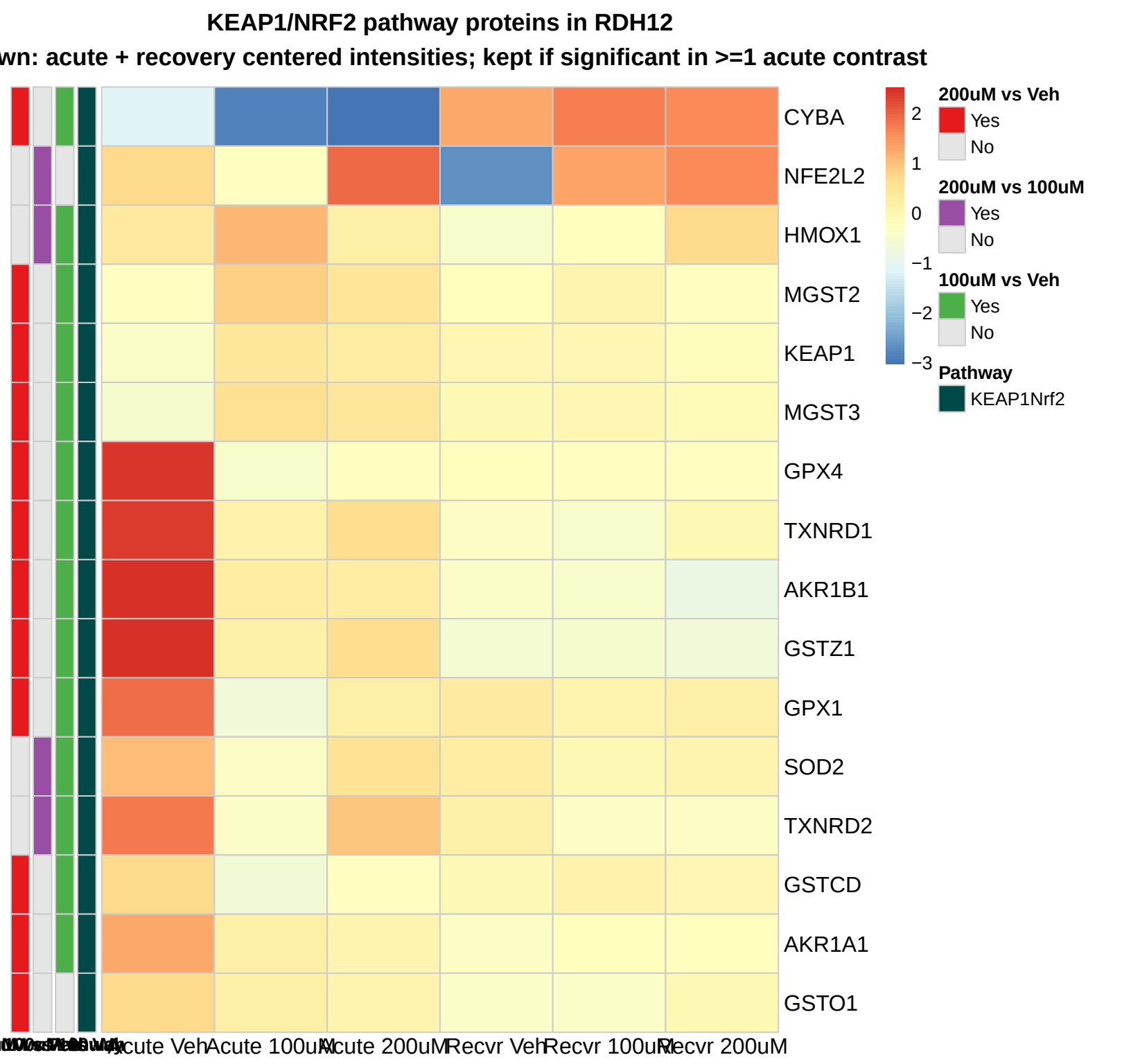

Acute RDH12 pathway proteins (sig in >=1 acute contrast)
Centered intensities; rows clustered within pathway groups
NFE2L2
TXNRD2
TXNRD1
AKR1B1
GSTZ1
GSTCD
AKR1A1
GSTO1
HMOX1
MGST2
KEAP1
MGST3
Acute Veh
Acute 100uM
Acute 200uM
NA
NA
NA
2
1
0
−1
−2
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2

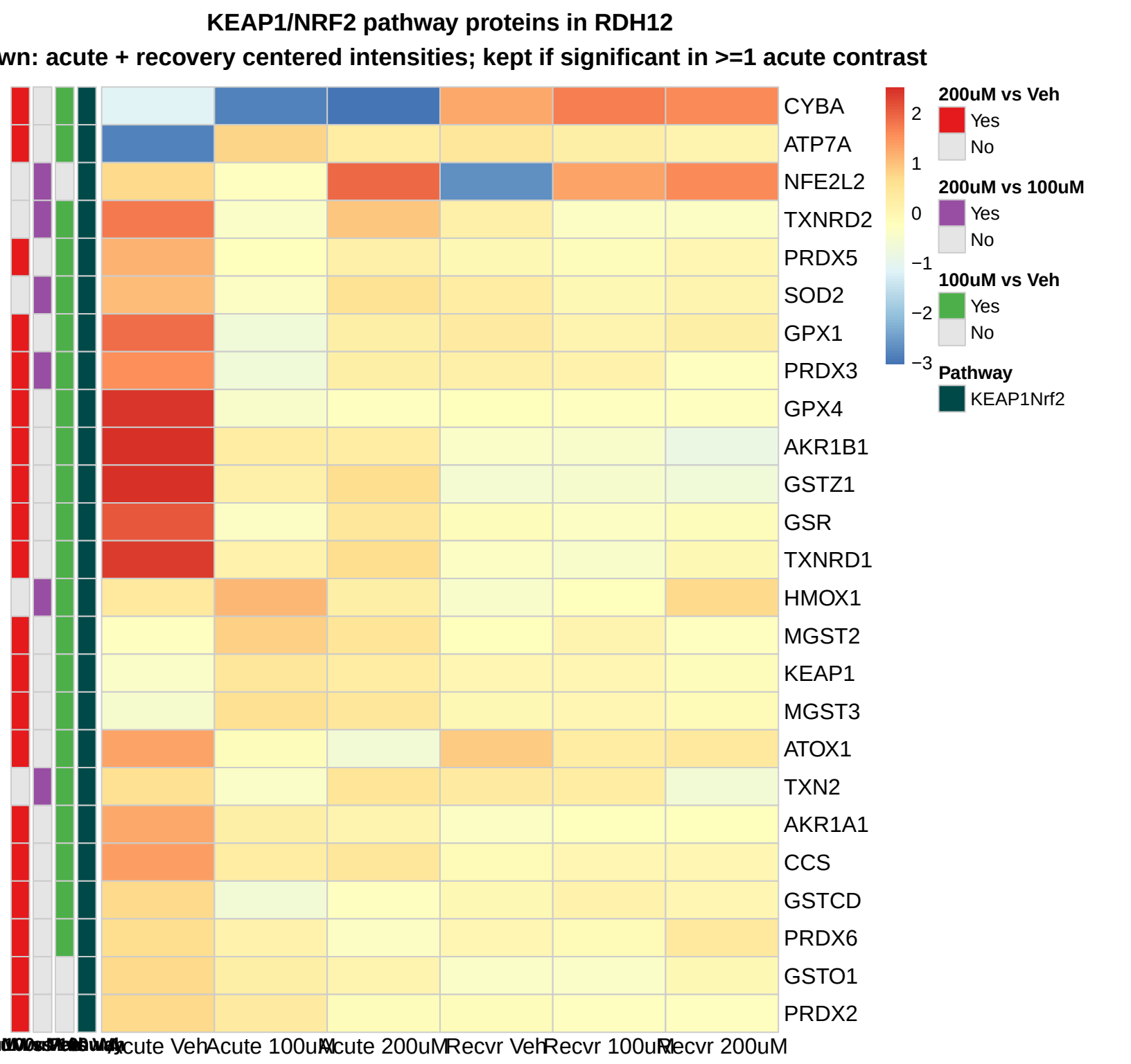

Acute RDH12 pathway proteins (sig in >=1 acute contrast)
Centered intensities; rows clustered within pathway groups
NFE2L2
TXNRD2
TXNRD1
AKR1B1
GSTZ1
GSTCD
AKR1A1
GSTO1
HMOX1
MGST2
KEAP1
MGST3
Acute Veh
Acute 100uM
Acute 200uM
NA
NA
NA
2
1
0
−1
−2
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2

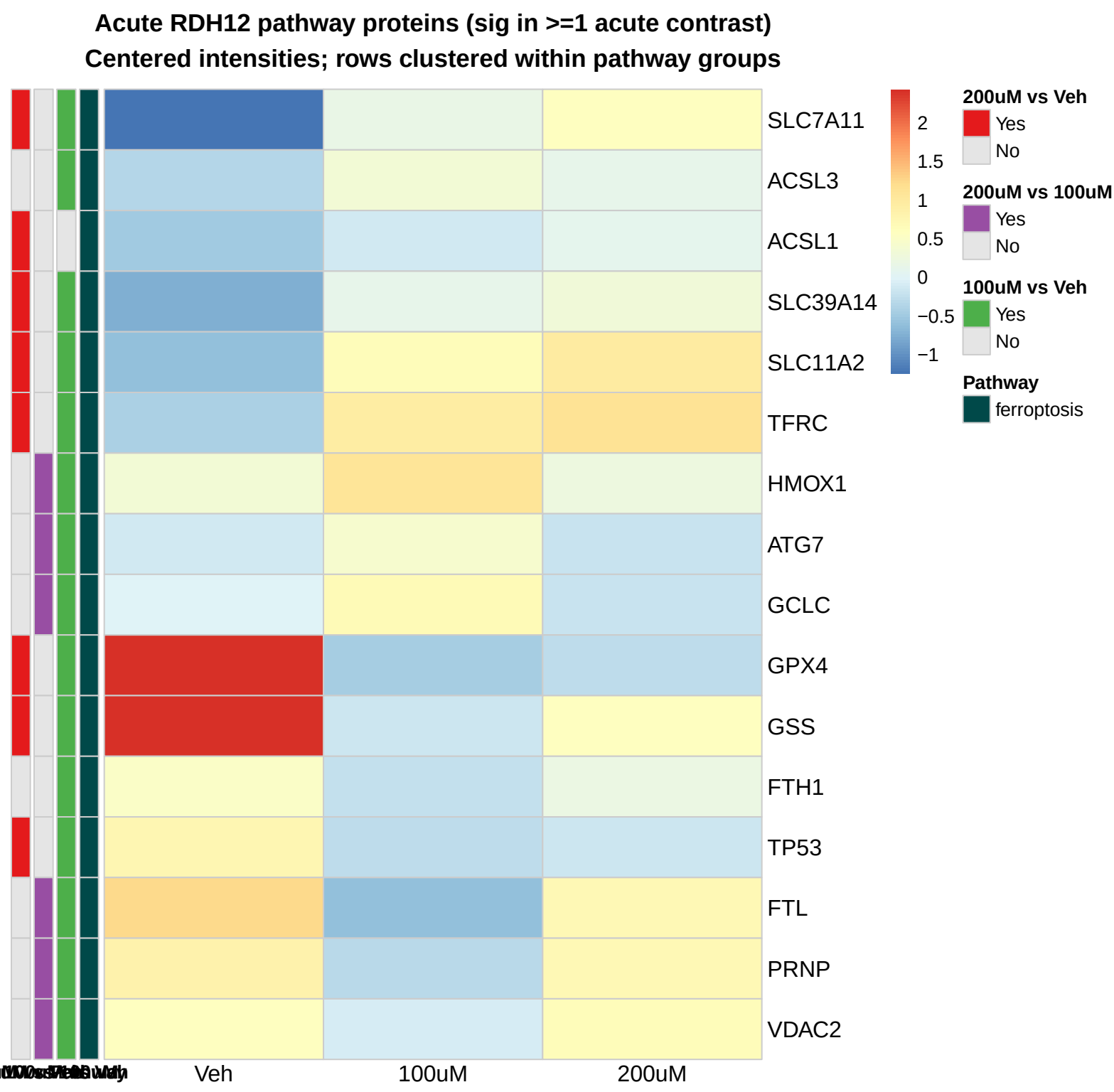

KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
NFE2L2
TXNRD2
TXNRD1
AKR1B1
GSTZ1
GSTCD
AKR1A1
GSTO1
HMOX1
MGST2
KEAP1
MGST3
2
1
0
−1
−2
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM


KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
NFE2L2
HMOX1
MGST2
KEAP1
MGST3
TXNRD1
AKR1B1
GSTZ1
SOD2
TXNRD2
GSTCD
AKR1A1
GSTO1
cute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2


KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
CYBA
NFE2L2
HMOX1
MGST2
KEAP1
MGST3
TXNRD1
AKR1B1
GSTZ1
SOD2
TXNRD2
GSTCD
AKR1A1
GSTO1
cute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2

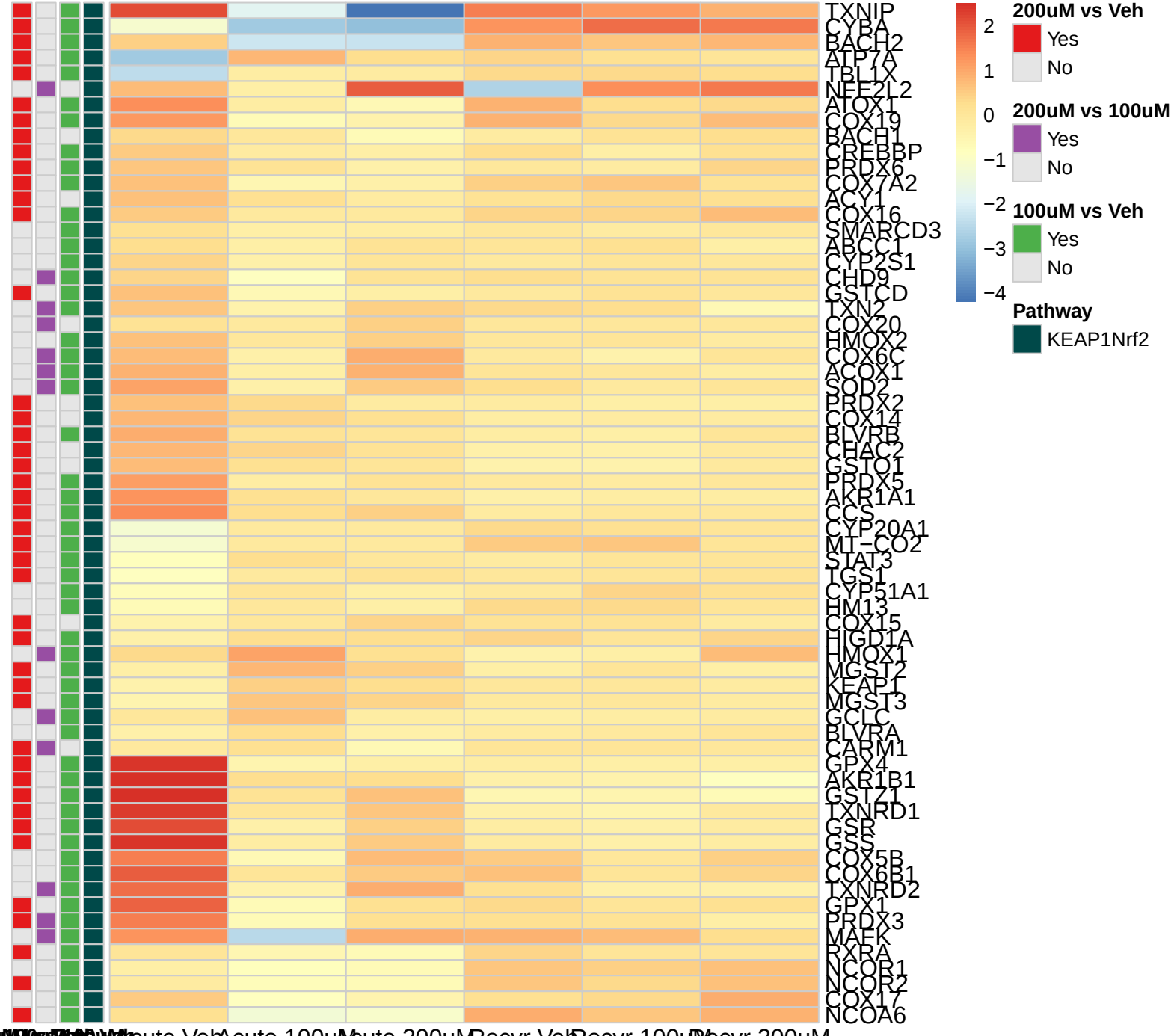

KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
CYBA
NFE2L2
HMOX1
MGST2
KEAP1
MGST3
GPX4
TXNRD1
AKR1B1
GSTZ1
GPX1
SOD2
TXNRD2
GSTCD
AKR1A1
GSTO1
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2


KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
CYBA
ATP7A
NFE2L2
TXNRD2
PRDX5
SOD2
GPX1
PRDX3
GPX4
AKR1B1
GSTZ1
GSR
TXNRD1
HMOX1
MGST2
KEAP1
MGST3
ATOX1
TXN2
AKR1A1
CCS
GSTCD
PRDX6
GSTO1
PRDX2
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2


Acute RDH12 pathway proteins (sig in >=1 acute contrast)
Centered intensities; rows clustered within pathway groups
SLC7A11
ACSL3
ACSL1
SLC39A14
SLC11A2
TFRC
HMOX1
ATG7
GCLC
GPX4
GSS
FTH1
TP53
FTL
PRNP
VDAC2
Veh
100uM
200uM
2
1.5
1
0.5
0
−0.5
−1
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
ferroptosis

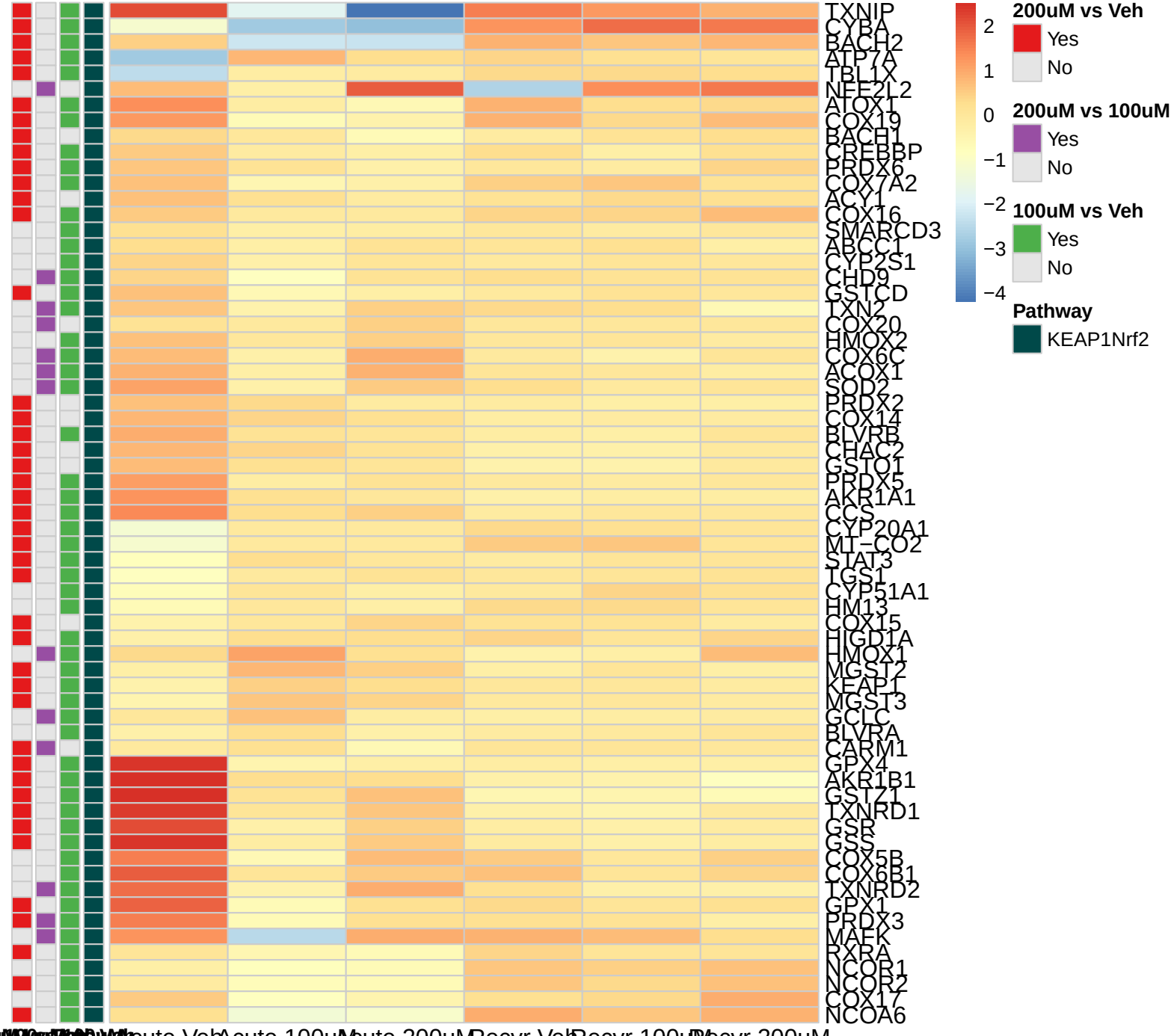
KEAP1/NRF2 pathway proteins in RDH12

wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast



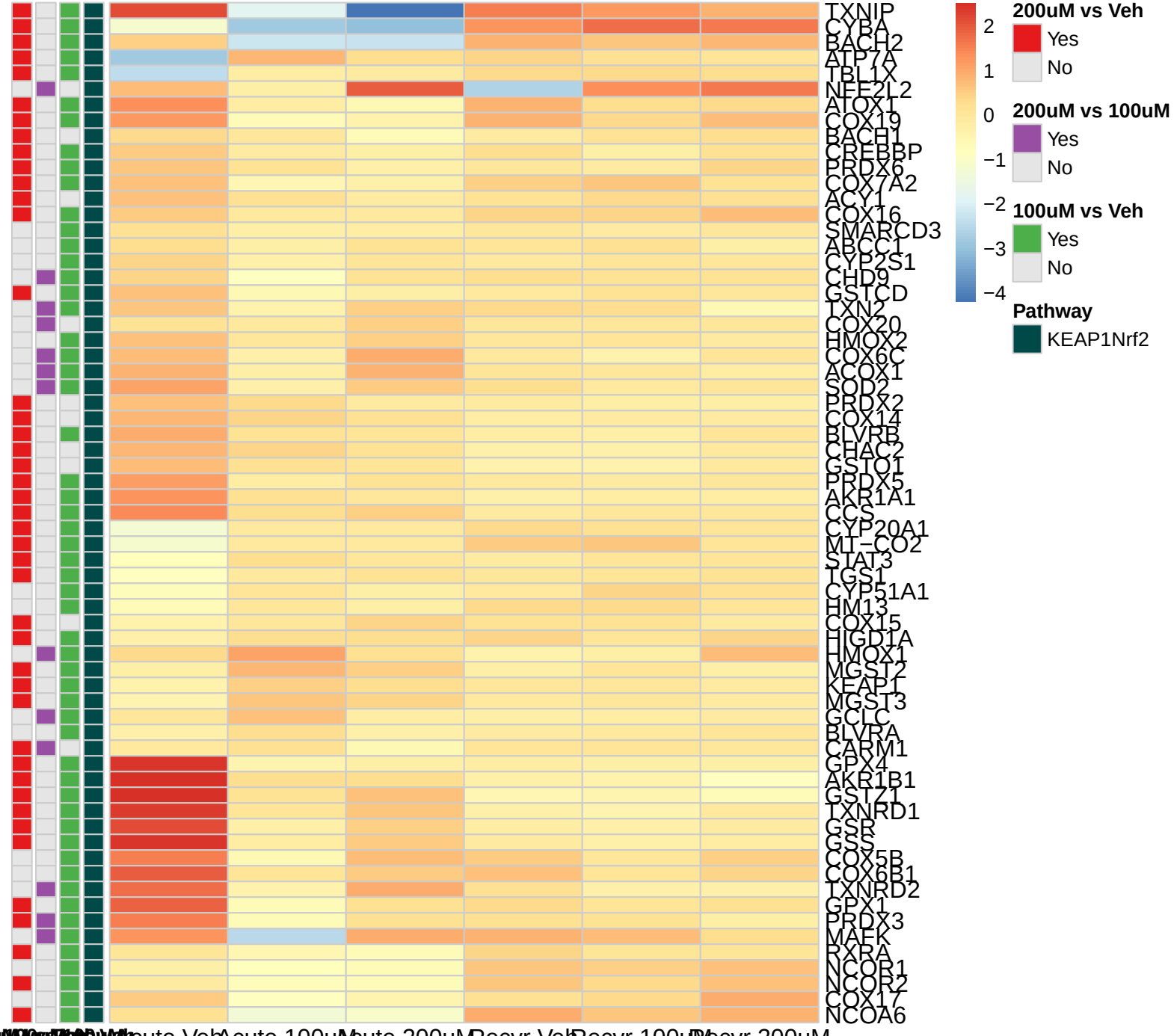
KEAP1/NRF2 pathway proteins in RDH12

wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast



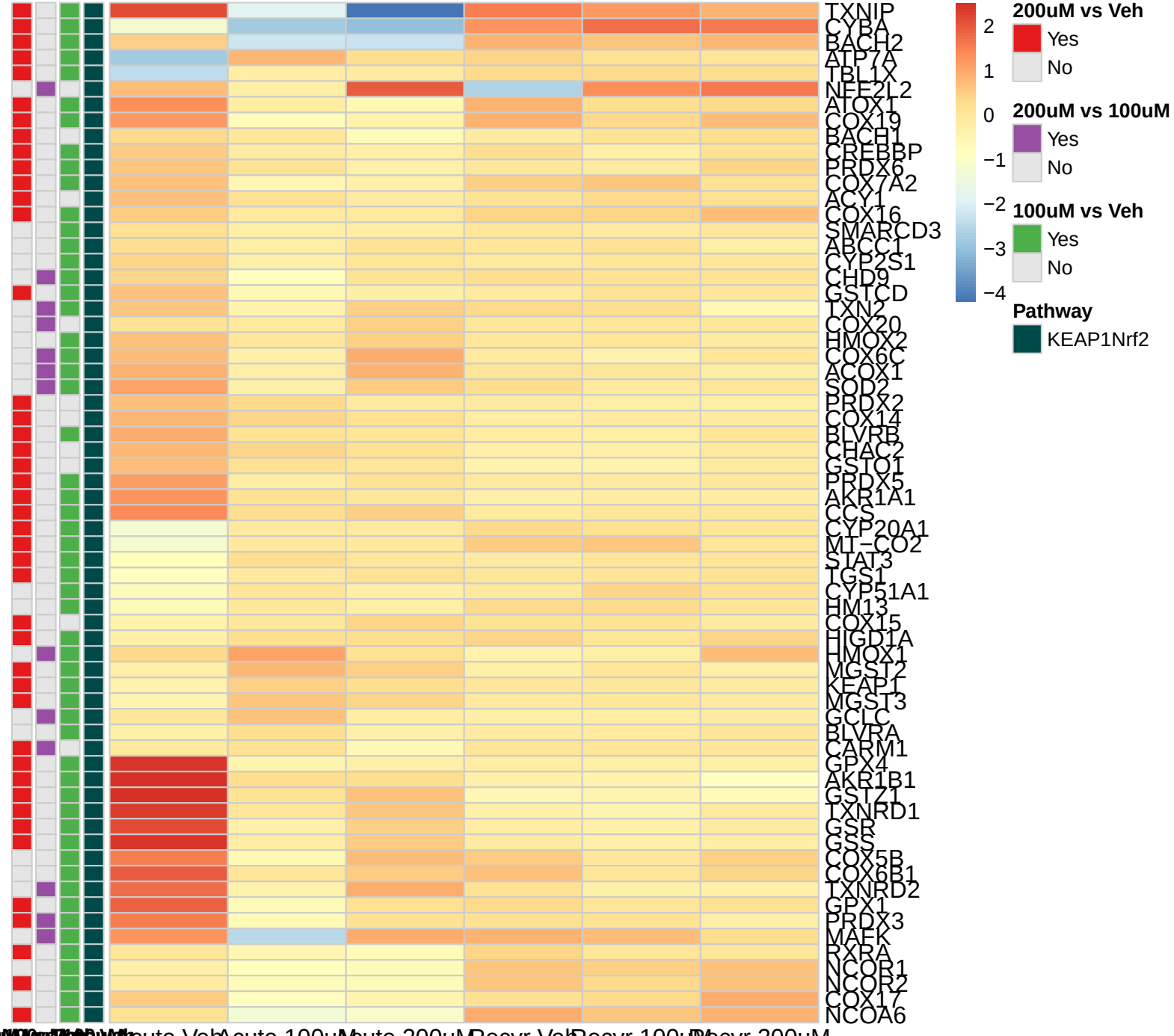
KEAP1/NRF2 pathway proteins in RDH12

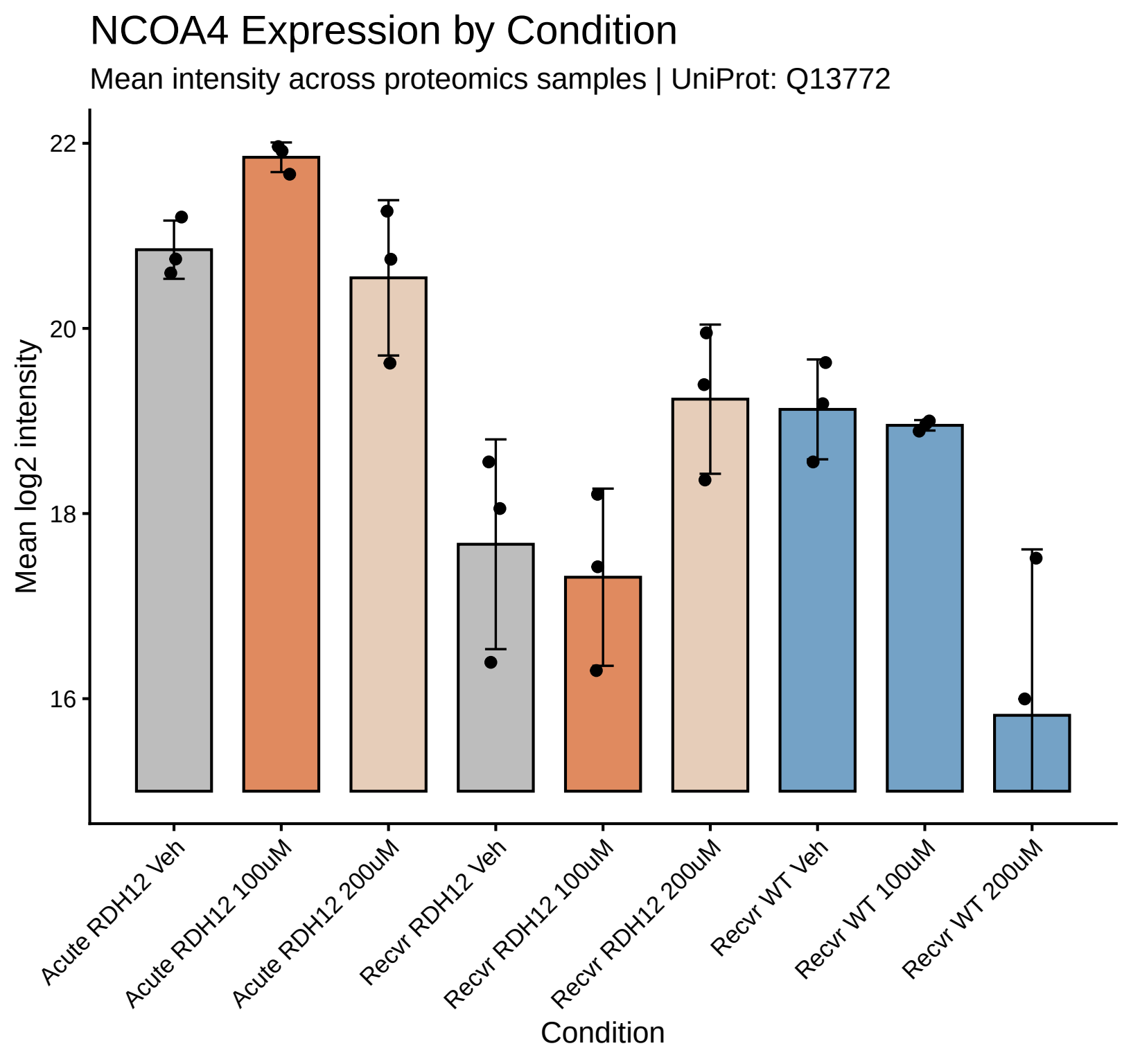
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast

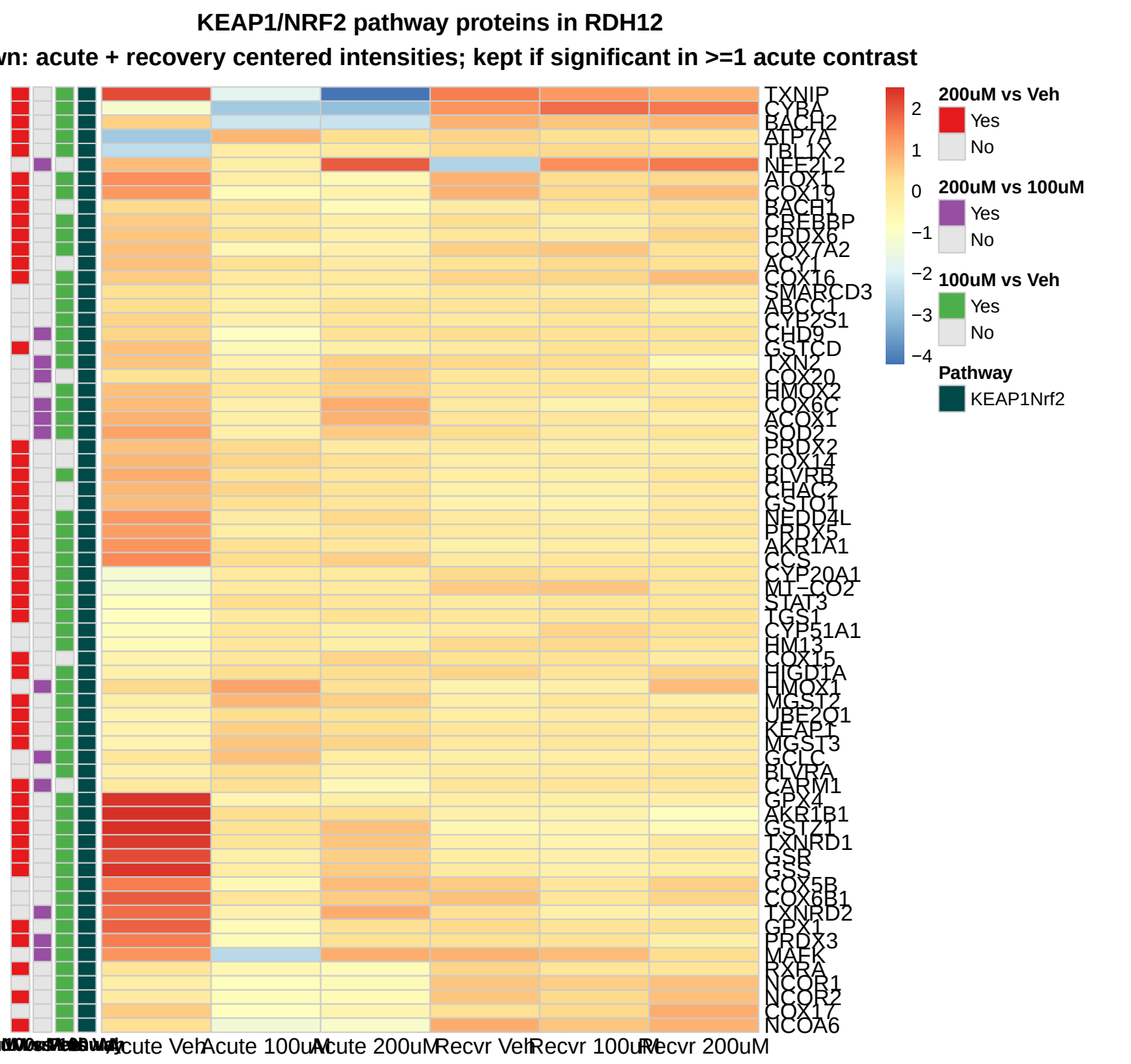


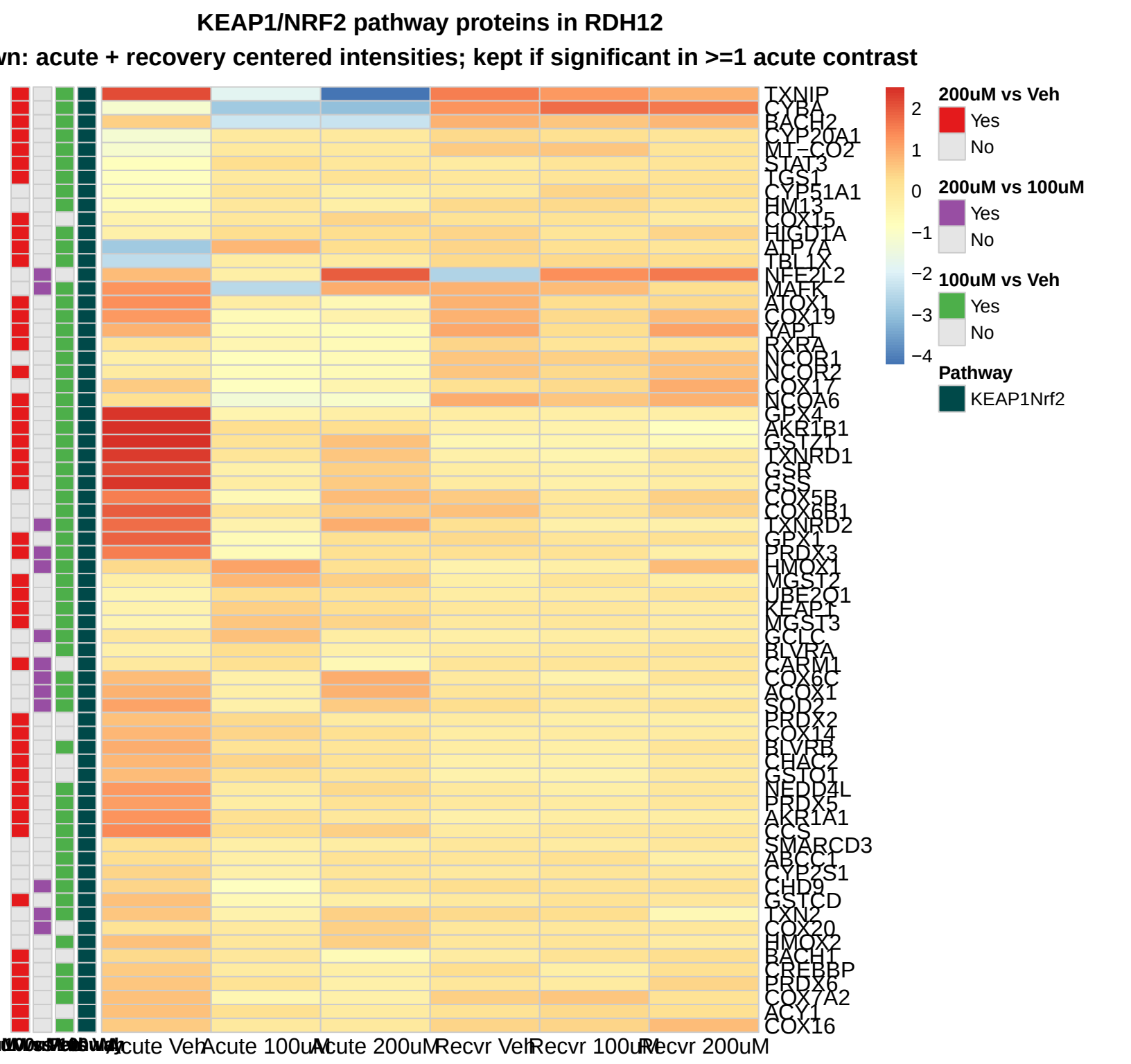
KEAP1/NRF2 pathway proteins in RDH12

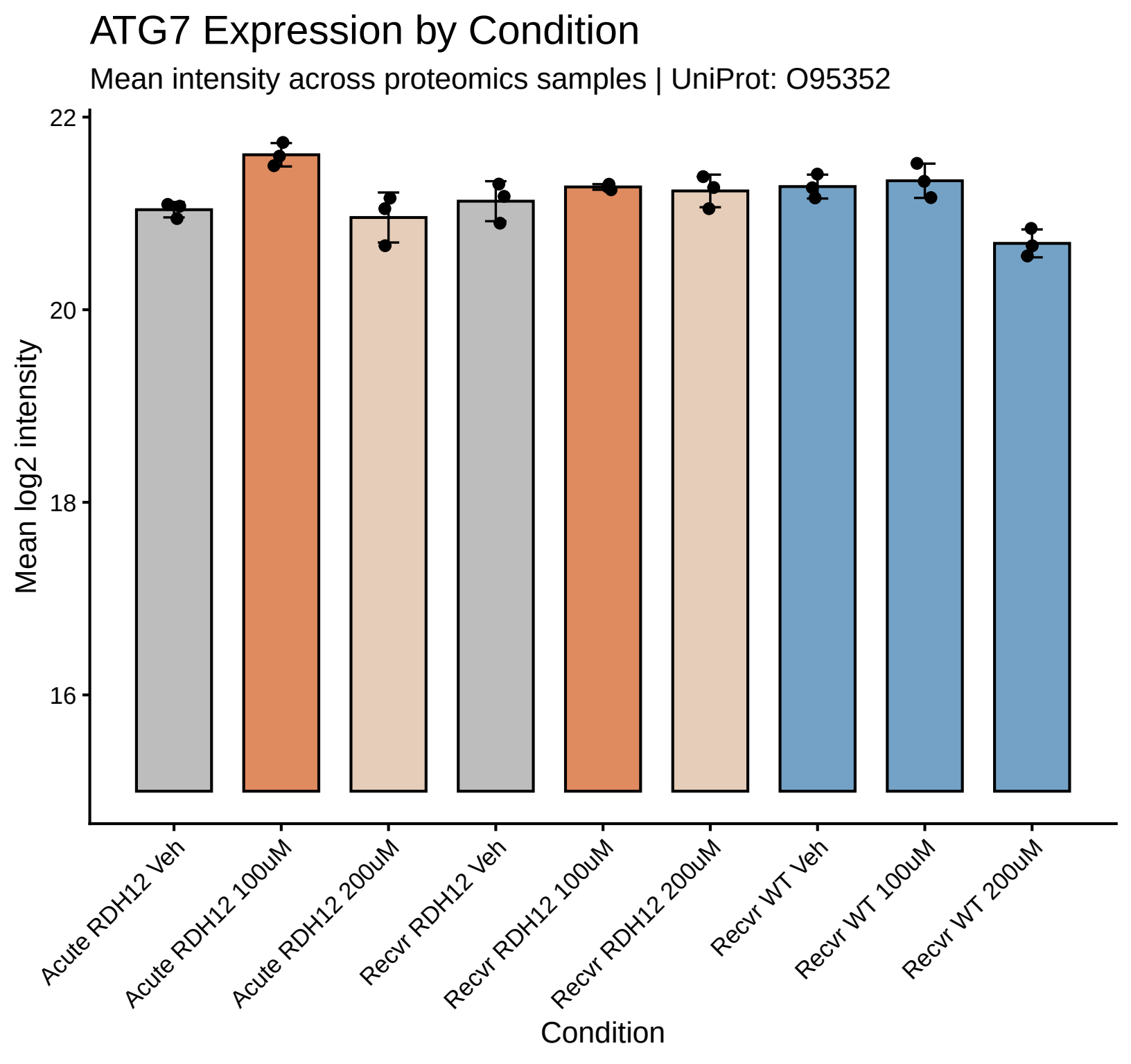
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast

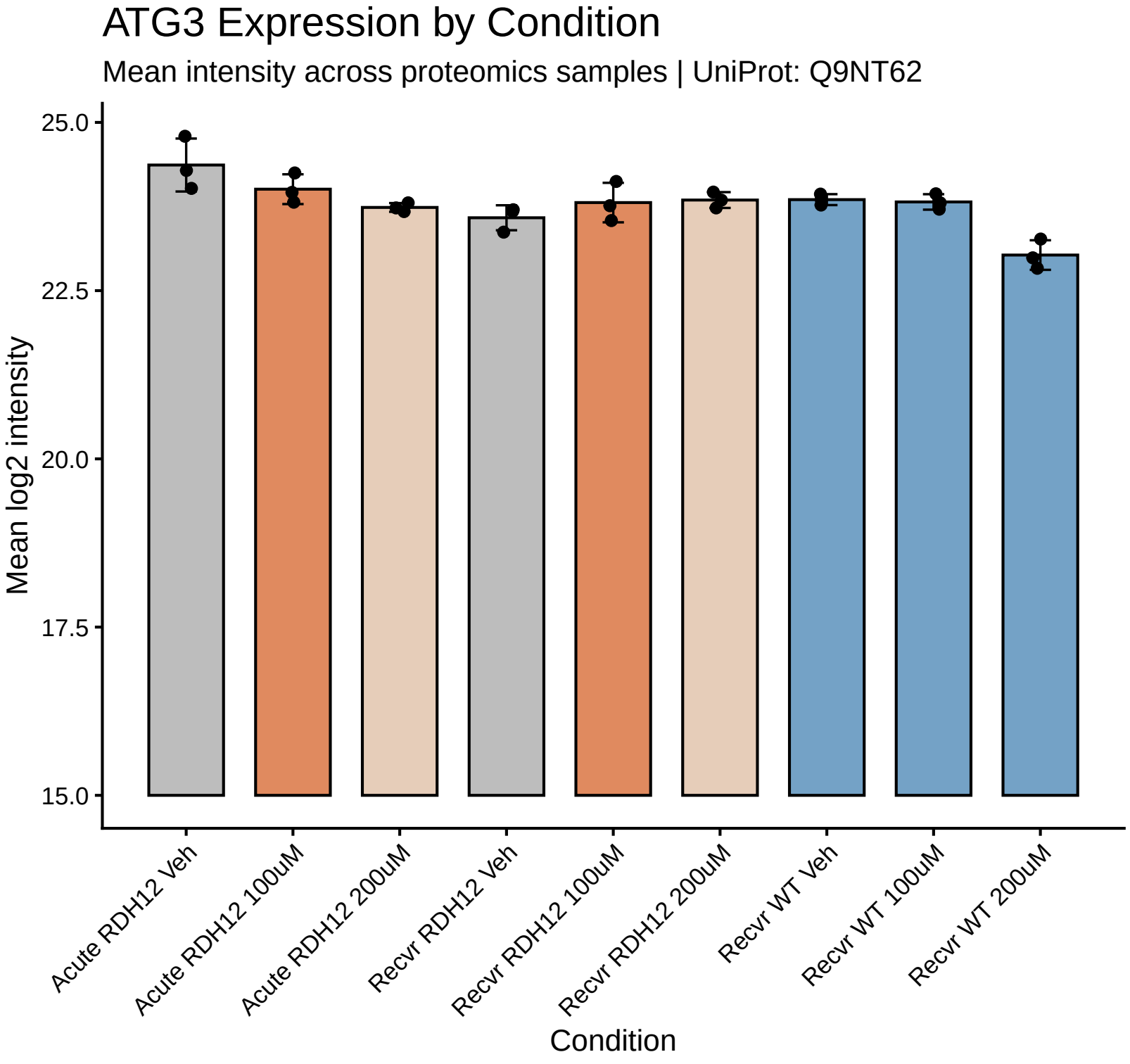


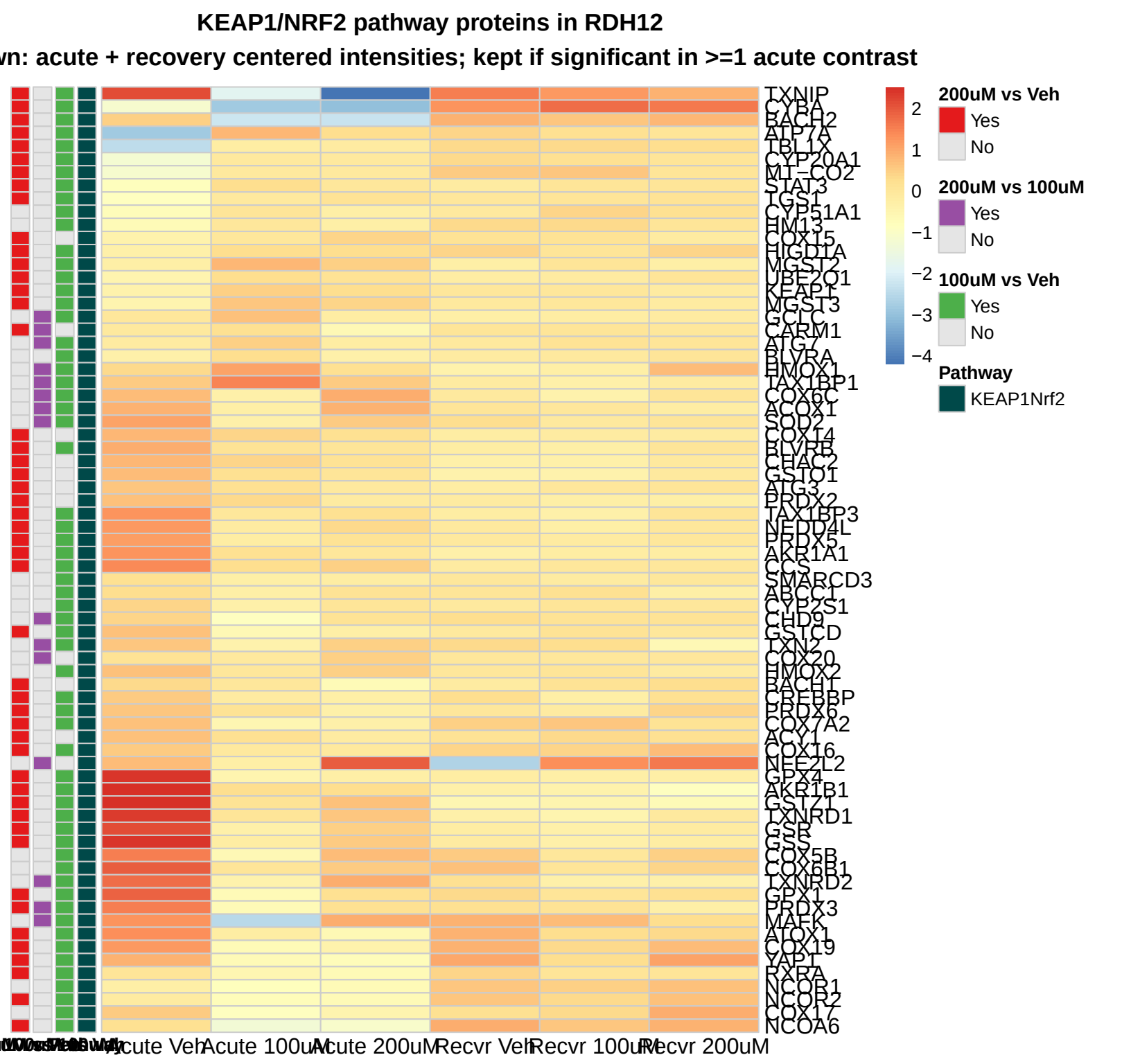

NCOA4 Expression by Condition
Mean intensity across proteomics samples | UniProt: Q13772
Mean log2 intensity
22
20
18
16
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition

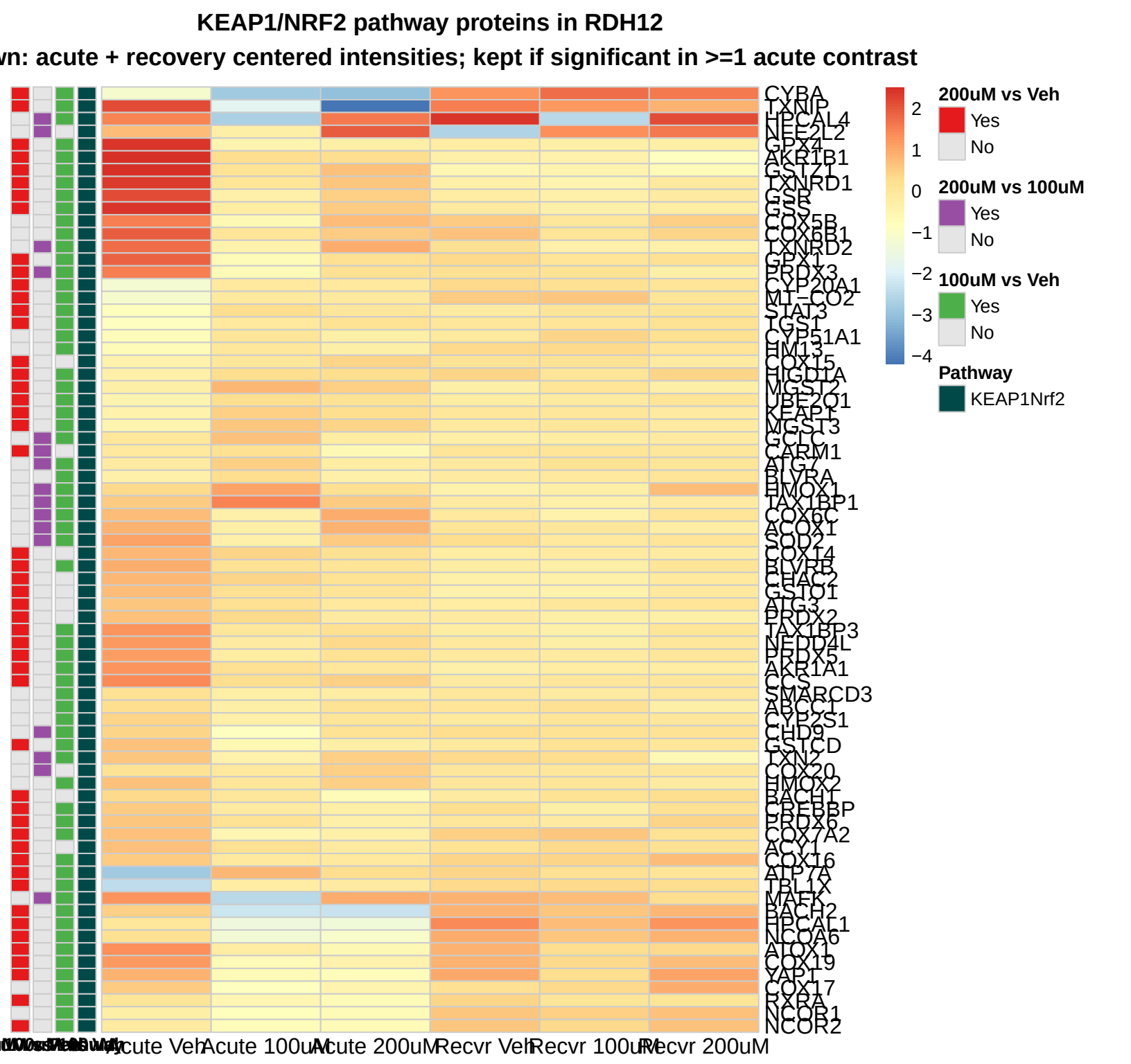

KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
TXNIP
CYBA
BACH2
ATP7A
TBL1X
NFE2L2
ATOX1
COX19
BACH1
CREBBP
PRDX6
COX7A2
ACY1
COX16
SMARCD3
ABCC1
CYB5S1
CHD9
GSTCD
TXN2
COX20
HMOX2
COX6C
ACOX1
SOD2
PRDX2
COX14
BLVRB
CHAC2
GSTO1
NEDD4L
PRDX5
AKR1A1
CCS
CYP20A1
MT−CO2
STAT3
TGS1
CYP51A1
HM13
COX15
HIGD1A
HMOX1
MGST2
UBE2Q1
KEAP1
MGST3
GCLC
BLVRA
CARM1
GPX4
AKR1B1
GSTZ1
TXNRD1
GSR
GSS
COX5B
COX6B1
TXNRD2
GPX1
PRDX3
MAFK
RXRA
NCOR1
NCOR2
COX17
NCOA6
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
−4
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2

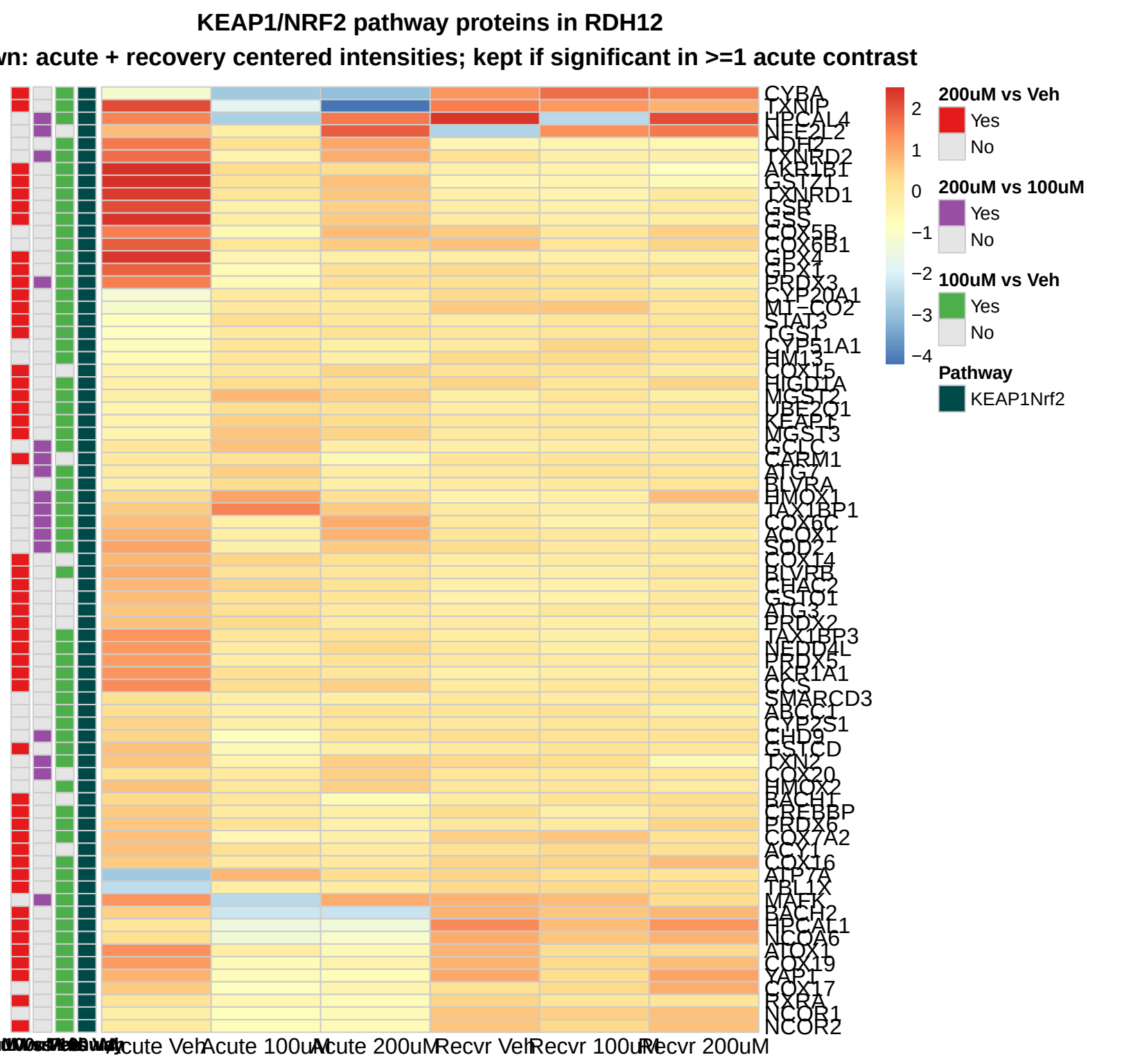

KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
TXNIP
CYBA
BACH2
CYP20A1
MT−CO2
STAT3
TGS1
CYP51A1
HM13
COX15
HIGD1A
ATP7A
TBL1X
NFE2L2
MAFK
ATOX1
COX19
YAP1
RXRA
NCOR1
NCOR2
COX17
NCOA6
GPX4
AKR1B1
GSTZ1
TXNRD1
GSR
GSS
COX5B
COX6B1
TXNRD2
GPX1
PRDX3
HMOX1
MGST2
UBE2Q1
KEAP1
MGST3
GCLC
BLVRA
CARM1
COX6C
ACOX1
SOD2
PRDX2
COX14
BLVRB
CHAC2
GSTO1
NEDD4L
PRDX5
AKR1A1
CCS
SMARCD3
ABCC1
CYP2S1
CHD9
GSTCD
TXN2
COX20
HMOX2
BACH1
CREBBP
PRDX6
COX7A2
ACY1
COX16
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
−4
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2

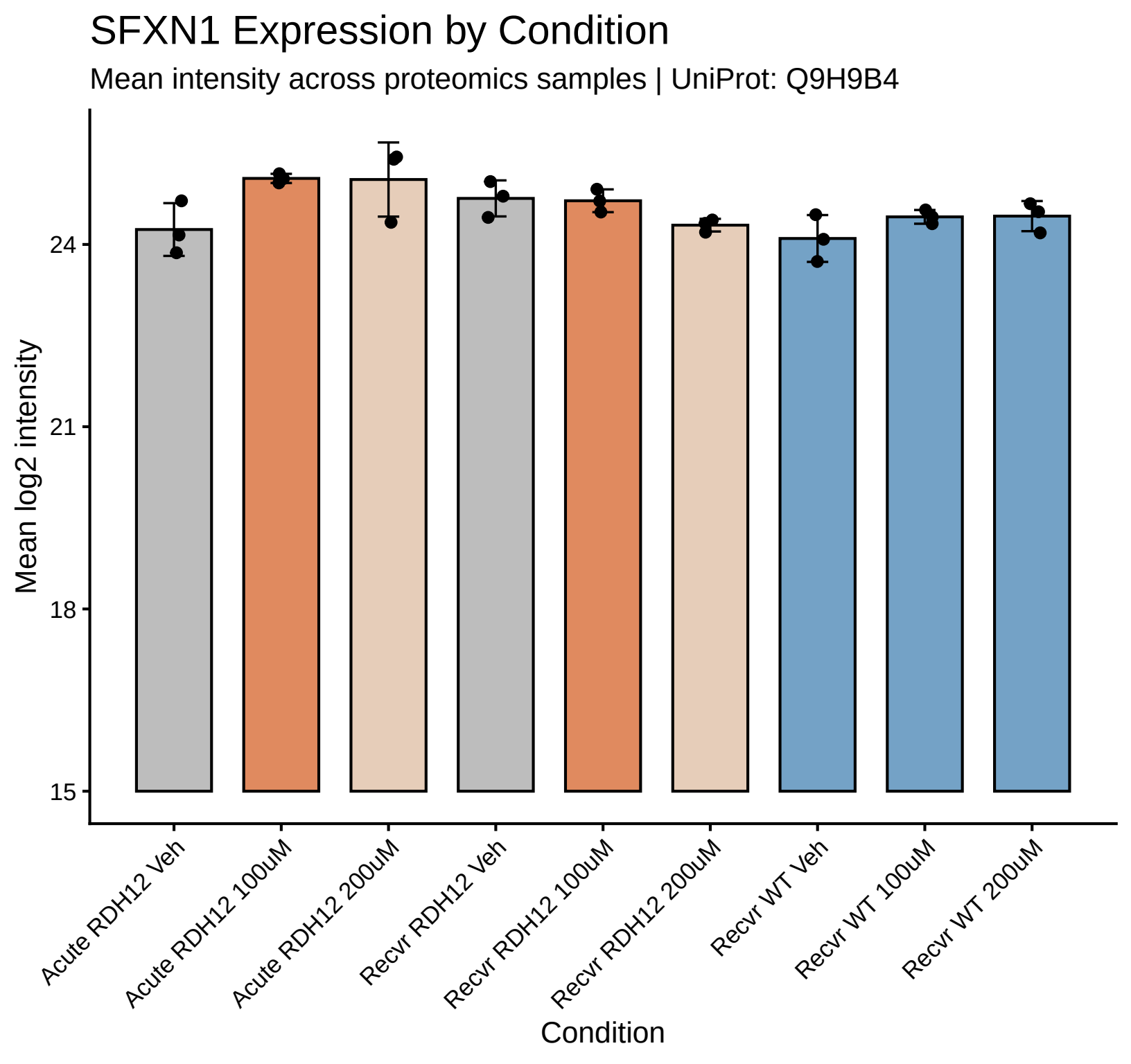

ATG7 Expression by Condition
Mean intensity across proteomics samples | UniProt: O95352
Mean log2 intensity
22
20
18
16
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition

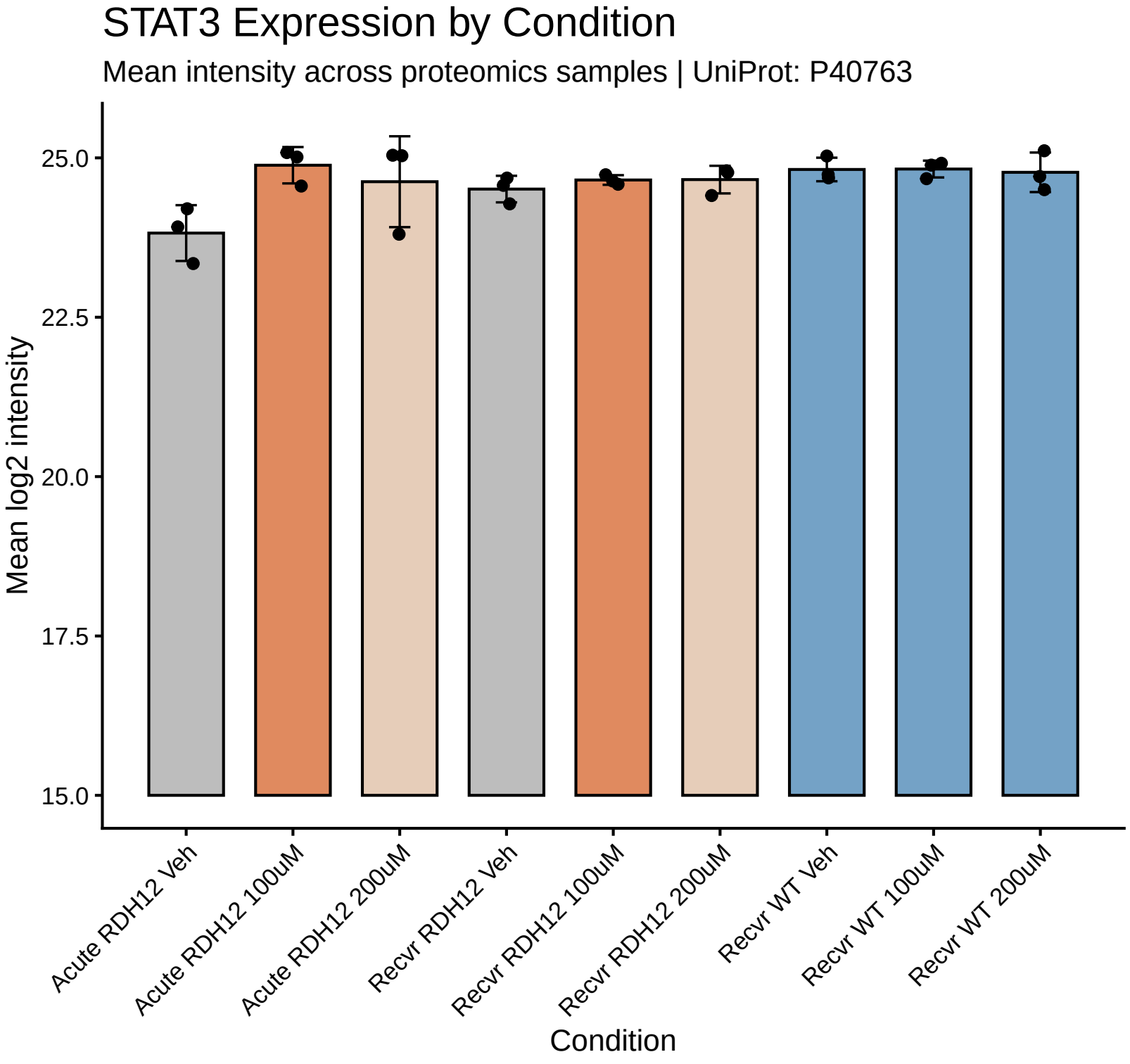

ATG3 Expression by Condition
Mean intensity across proteomics samples | UniProt: Q9NT62
Mean log2 intensity
25.0
22.5
20.0
17.5
15.0
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition

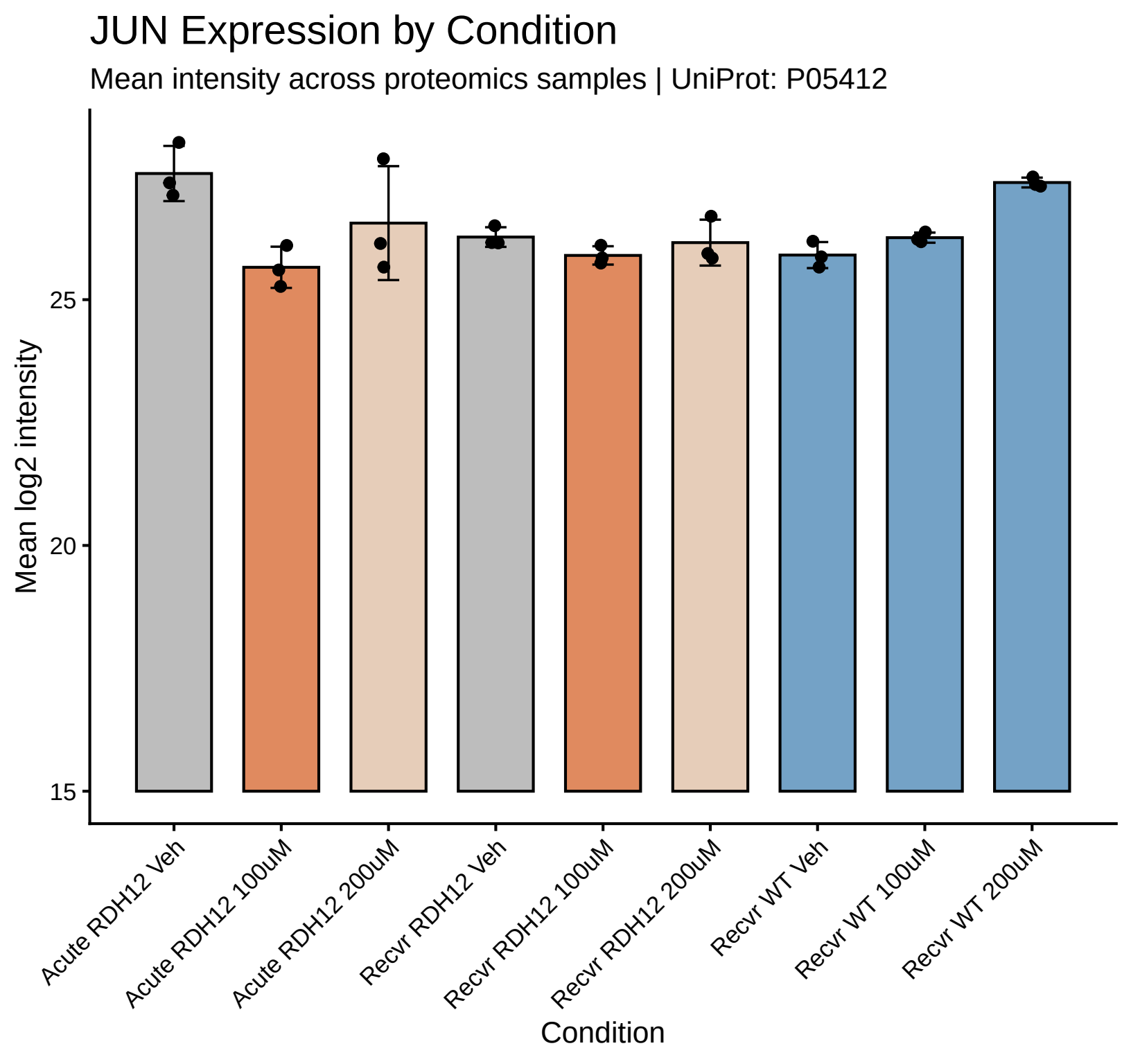

KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
TXNIP
CYBA
BACH2
ATP7A
TBL1X
CYP20A1
MT-CO2
STAT3
TGS1
CYP51A1
HM13
COX15
HIGD1A
MGST2
UBE2Q1
KEAP1
MGST3
GCLC
CARM1
ATG7
BLVRA
HMOX1
TAX1BP1
COX6C
ACOX1
SOD2
COX14
BLVRB
CHAC2
GSTO1
ATG3
PRDX2
TAX1BP3
NEDD4L
PRDX5
AKR1A1
CCS
SMARCD3
ABCC1
CYP2S1
CHD9
GSTCD
TXN2
COX20
HMOX2
BACH1
CREBBP
PRDX6
COX7A2
ACY1
COX16
NFE2L2
GPX4
AKR1B1
GSTZ1
TXNRD1
GSR
GSS
COX5B
COX6B1
TXNRD2
GPX1
PRDX3
MAFK
ATOX1
COX19
YAP1
RXRA
NCOR1
NCOR2
COX17
NCOA6
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
−4
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2

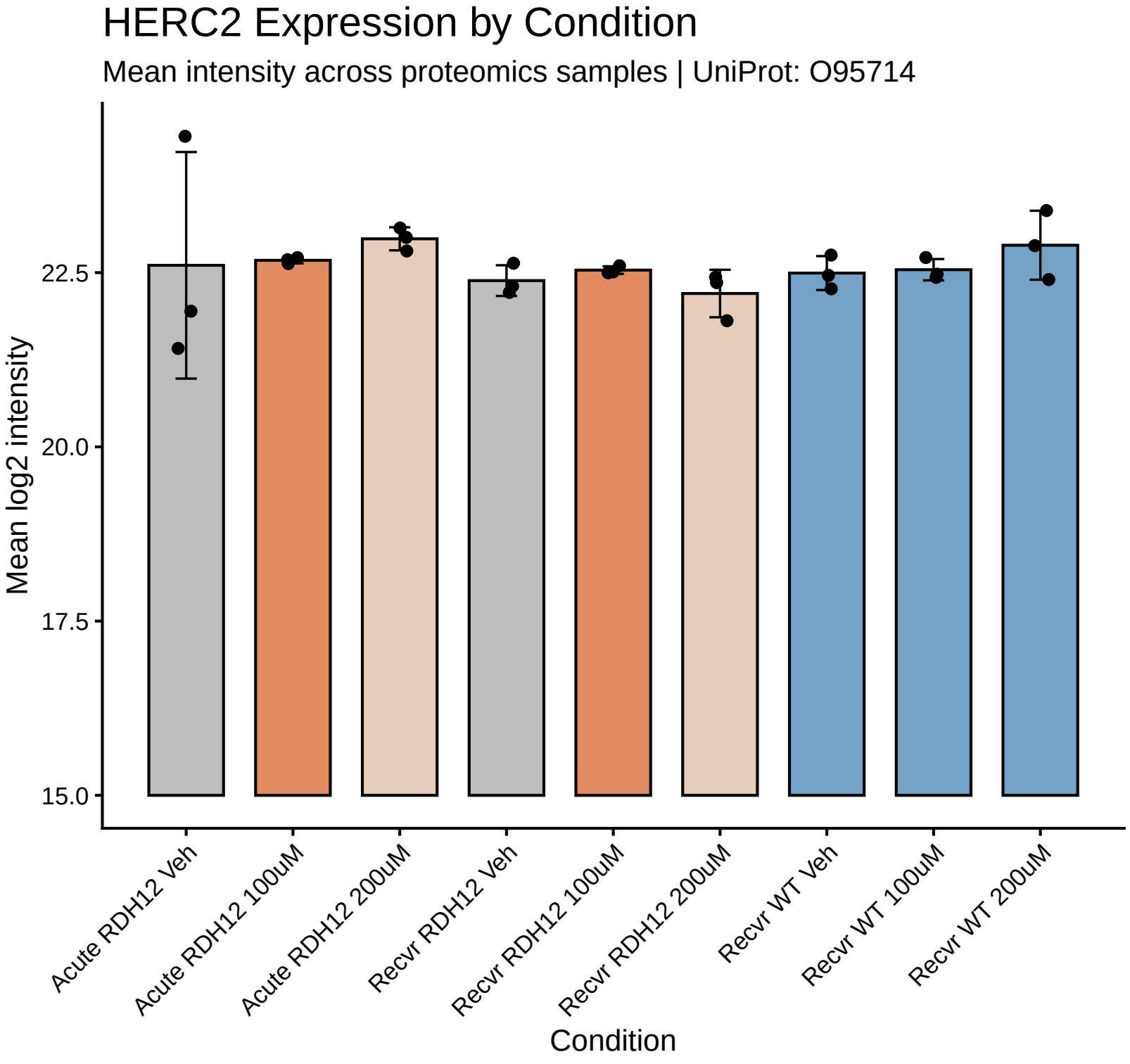

KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
CYBA
TXNIP
HPCAL4
NFE2L2
GPX4
AKR1B1
GSTZ1
TXNRD1
GSR
GSS
COX5B
COX6B1
TXNRD2
GPX1
PRDX3
CYP20A1
MT-CO2
STAT3
TGS1
CYP51A1
HM13
COX15
HIGD1A
MGST2
UBE2Q1
KEAP1
MGST3
GCLC
CARM1
ATG7
BLVRA
HMOX1
TAX1BP1
COX6C
ACOX1
SOD2
COX14
BLVRB
CHAC2
GSTO1
ATG3
PRDX2
TAX1BP3
NEDD4L
PRDX5
AKR1A1
CCS
SMARCD3
ABCC1
CYP2S1
CHD9
GSTCD
TXN2
COX20
HMOX2
BACH1
CREBBP
PRDX6
COX7A2
ACY1
COX16
ATP7A
TBL1X
MAFK
BACH2
HPCAL1
NCOA6
ATOX1
COX19
YAP1
COX17
RXRA
NCOR1
NCOR2
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
−4
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2


KEAP1/NRF2 pathway proteins in RDH12
wn: acute + recovery centered intensities; kept if significant in >=1 acute contrast
CYBA
TXNIP
HPCAL4
NFE2L2
CDH2
TXNRD2
AKR1B1
GSTZ1
TXNRD1
GSR
GSS
COX5B
COX6B1
GPX4
GPX1
PRDX3
CYP20A1
MT-CO2
STAT3
TGS1
CYP51A1
HM13
COX15
HIGD1A
MGST2
UBE2Q1
KEAP1
MGST3
GCLC
CARM1
ATG7
BLVRA
HMOX1
TAX1BP1
COX6C
ACOX1
SOD2
COX14
BLVRB
CHAC2
GSTO1
ATG3
PRDX2
TAX1BP3
NEDD4L
PRDX5
AKR1A1
CCS
SMARCD3
ABCC1
CYP2S1
CHD9
GSTCD
TXN2
COX20
HMOX2
BACH1
CREBBP
PRDX6
COX7A2
ACY1
COX16
ATP7A
TBL1X
MAFK
BACH2
HPCAL1
NCOA6
ATOX1
COX19
YAP1
COX17
RXRA
NCOR1
NCOR2
Acute Veh
Acute 100uM
Acute 200uM
Recvr Veh
Recvr 100uM
Recvr 200uM
2
1
0
−1
−2
−3
−4
200uM vs Veh
Yes
No
200uM vs 100uM
Yes
No
100uM vs Veh
Yes
No
Pathway
KEAP1Nrf2


SFXN1 Expression by Condition
Mean intensity across proteomics samples | UniProt: Q9H9B4
Mean log2 intensity
24
21
18
15
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition


STAT3 Expression by Condition
Mean intensity across proteomics samples | UniProt: P40763
Mean log2 intensity
25.0
22.5
20.0
17.5
15.0
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition


JUN Expression by Condition
Mean intensity across proteomics samples | UniProt: P05412
Mean log2 intensity
25
20
15
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition


HERC2 Expression by Condition
Mean intensity across proteomics samples | UniProt: O95714
Mean log2 intensity
22.5
20.0
17.5
15.0
Acute RDH12 Veh
Acute RDH12 100uM
Acute RDH12 200uM
Recvr RDH12 Veh
Recvr RDH12 100uM
Recvr RDH12 200uM
Recvr WT Veh
Recvr WT 100uM
Recvr WT 200uM
Condition